Gayathri Krishnamoorthy

Electrical Engineer| Computer Scientist

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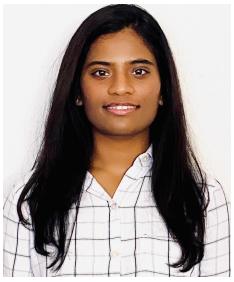

Power Systems Engineer with 4+ years of experience as a researcher at Washington State University and multiple internship experiences at National Laboratories. Worked on several projects developing software tools to efficiently integrate renewable resources into power grid models while making them a significant player in energy markets. I am outgoing, creative, and detail-oriented with strong engineering basics and communication skills. I am effective both solo or on a team. In my spare time I practice yoga, meditation, and painting. I am also passionate about the brain science behind mindfulness practices and keeping up with recent developments in AI.

Education

Present – August 2018 | **Ph.D. in Electrical Engineering , Washington State University, Pullman, WA**

- Research : Frequency Regulation Market Service Provision from Distribution Connected Battery Energy Storage Systems
- Major : Power Systems
- Minor : Computer Science
- Advisor : Dr. Anamika Dubey (WSU)
- Co-Advisors : Dr. Anjan Bose (WSU), Dr. Mani Venkatasubramanian (WSU), and Dr. Laurentiu Marinovici (PNNL)

May 2018 – Aug 2016 | **M.S. in Electrical Engineering, Washington State University, Pullman, WA**

- Research : An Iterative Co-simulation Framework for the Integrated Transmission and Distribution System Analysis
- Major : Power Systems
- Minor : Computer Engineering
- Advisor : Dr. Anamika Dubey (WSU)

May 2016 – Aug 2012 | **B.E. in Electronics and Communications Engineering, Anna University, Chennai, India**

- Title : Smart Home Design with Voice Recognition in a Single Controller (Arduino Uno) Environment
- Major : Embedded Systems
- Minor : Communications
- Advisor : Dr. Suresh T

Professional Experience

Present – August 2016 | **Graduate Research Assistant, Washington State University, Pullman, WA**

- Developing efficient regulation market techniques to enable transmission level frequency regulation service provision using distribution connected battery energy storage systems *(Python)*.
- Developing model-based and model-free deep reinforcement learning techniques to mitigate power system level operational violations with the increasing integration of distributed energy resources *(Python, Tensorflow, Keras)*.
- Designed and developed a transmission and distribution (T&D) co-simulation platform to understand the impacts of increasing renewable penetrations on power system operations *(MATLAB, Python, and OpenDSS)*.
- Developed an optimal retail market model and performed impact assessment of demand-side renewable resource participation and its integration into the wholesale market *(MATLAB, OpenDSS)*.

Python | MATLAB | OpenDSS | PSAT | Tensorflow | Keras

Dec 2019
Jan 2019

Graduate Teaching Assistant, Washington State University, Pullman, WA

Courses : Numerical Computing for Engineers, Professional Skills in Computing and Engineering

Tasks :

- Engaged classes of 100+ undergraduate (multiple engineering discipline) students
- Helped the class in learning to use MATLAB (a professional computing software) and Microsoft project professional softwares
- Provided my best efforts in evaluating students' homeworks and exams

MATLAB | Microsoft Project Professional | LaTex

Dec 2018
May 2018

Graduate Research Intern, Pacific Northwest National Laboratory, Seattle, WA

- Developed deep reinforcement learning models to perform AC optimal power flow for the power transmission system optimization.
- Assisted in development of a transactive energy software agent, TESP, that optimizes the dynamics of distributed energy resources in a household and bids the excess energy in the energy market to provide monetary benefits to the costumers using electric vehicles, batteries, and PVs.
- Performed scalability testing for the open source co-simulation platform (HELICS) developed at PNNL in multiple programming language environments.
- HVAC market participation simulations in day-ahead and real-time markets using Transactive energy simulation platform (TESP) with HELICS on a Linux environment
- Developed docker environment for a microgrid restoration application with FNCS.

C++ | Python | MATLAB | MATPOWER | GridLAB-D | OpenDSS | Docker | EnergyPlus | FNCS | TESP | HELICS

Projects

Frequency Regulation Service Provision from Distribution Connected Energy Storage Systems — 2019-Present

github.com/gayukrishna/DRL-for-Secondary-Control-Services

In this project, I develop an efficient control model to utilize the flexibility of distribution connected battery energy storage systems in providing transmission frequency regulation services. This involves designing an optimal regulation market model at the distribution level that satisfies system level constraints while providing monetary benefits to the market participants

MATLAB | OpenDSS | Python

Model-based and Model-free Distribution System Optimization using Deep Reinforcement Learning — 2020

github.com/gayukrishna/DRL-for-Secondary-Control-Services

The goal is to develop an optimization strategy (model-based) and DRL techniques (model-free) to distribute and maintain the scheduled frequency regulation dispatch from the ISO and the distribution level network constraints. We also propose to use imitation learning based hybrid model to scale the reinforcement learning approach for larger systems.

Python | Tensorflow | Keras

L2RPN NEURIPS - Adaptability Track — 2020

github.com/gayukrishna/NIPS2020-WSU NeurIPS 2020

This challenge is the latest round in a series of power network control challenges organized by RTE France (electricity transmission system operator of France). The goal was to control electricity transport in power networks running closer to their operational limits while keeping people and equipment safe.

Grid2Op | OpenAI Gym | Python | Tensorflow

Transmission -Distribution Co-simulation — 2017 - 2019

github.com/WSU-DS/Cosimulation PSERC T-60

Developed a framework for the integrated T&D system analysis to understand the impacts of distribution-connected renewable resources (DERs) on transmission system operations and vice-versa (project funded by PSERC). Also, using the co-simulation framework developed, explored the utility of DER's in transmission system support.

MATLAB | OpenDSS | Python

Website for Distribution lab at WSU — 2019

DOPAL

Designed and maintain a professional website for my research group at WSU.

HTML | bootstrap | CSS | jquery

GMLC-TDC/ HELICS 2018

github.com/GMLC-TDC github.com/GMLC-TDC/MATPOWER-wrapper HELICS

HELICS is a coordinated effort by multiple National Labs to develop a platform that aids advanced power system operations. I worked on certain aspects of the development like scalability testing and wrapper development for multiple power system tools. One major takeaway during this internship was learning docker development.

C++ | Python | MATLAB | Matpower

TESP 2018

github.com/pnnl/tesp TESP

For this project, I was involved in developing the energy market structure (real-time and day-ahead) for HVAC (heating, ventilation and air conditioning) systems. The major tasks were to develop consensual bidding strategies and to test/validate on large scale power system networks.

Python | C++ | GridLAB-D | EnergyPlus

Publications

- 2019 G. Krishnamoorthy and A. Dubey, "Transmission–Distribution Cosimulation : Analytical Methods for Iterative Coupling", in IEEE Systems Journal.
- 2019 J. C. Bedoya, C. Liu, G. Krishnamoorthy and A. Dubey, "Bilateral Electricity Market in a Distribution System Environment", in IEEE Transactions on Smart Grid.
- 2019 Sen, P.K., Velaga, Y.N., Chen, A., Krishnamoorthy, G. and Dubey, A., 2019," Advancements in Co-Simulation Techniques in Combined Transmission & Distribution Systems Analysis," The Journal of Engineering.
- 2019 G. Krishnamoorthy, A. Dubey and P. K. Sen, "Iteratively-Coupled Co-simulation Framework for Unbalanced Transmission-Distribution System," 2019 IEEE Milan PowerTech, Milan, Italy, 2019, pp. 1-6.
- 2018 Krishnamoorthy, Gayathri, and Anamika Dubey. "A framework to analyze interactions between transmission and distribution systems." In 2018 IEEE Power & Energy Society General Meeting (PESGM), pp. 1-5. IEEE, 2018.

Programming Languages

Python	●●●●●
C++	●●●●●
MATLAB	●●●●●
C	●●●●●
HTML	●●●●○
Javascript	●●●○○
CSS	●●●○○

Simulation Tools

- Tensorflow
- Docker
- OpenDSS
- GridLAB-D
- HELICS
- OpenAI Gym
- Grid2Op

Professional Activities

- Present Serving as a reviewer for IEEE Power and Energy Society Conferences, Institution of Engineering and Technology Journals, and IEEE Systems Journal
- 2020 Participated in WCCI and NeurIPS 2020 learning to run power network challenges
- 2019 Authored a Ph.D. Research Proposal to Pacific Northwest National Laboratory
- 2019, 2018 Paper Presentations- IEEE Power and Energy Society General Meeting
- 2019, 2018, 2017 Poster Presentations - Power Systems Engineering Research Center
- 2015 Trained and participated in FreescaleCup Intelligent Car Racing Competition

Honors and Awards

- 2019 **Awardee** Graduate & Professional Student Association Scholarship for the Research Exposition at WSU
- 2018 **2nd Place** Best Student Paper Award at North American Power Symposium
- 2015 **Awardee** Completed course in CCNA Exploration : Network Fundamentals
- 2014 **Awardee** Level 1 Cambridge English Language Assessment in Business English

References

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